Shrividya Hegde

AI Engineer | LLM & RAG Systems | Data Orchestration (Apache Airflow)

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SUMMARY

AI engineer with 12 years in software and data engineering, focused on production data platforms, orchestration and LLM systems. Apache Airflow Champion (#1 globally, Q3 2026) with 7 merged pull requests to Apache Airflow. Partners with product, platform and client stakeholders to scope ambiguous problems, ship working integrations, and explain tradeoffs to technical and non-technical audiences.

TECHNICAL SKILLS

LLM & AI: RAG, LangChain, OpenAI API, Amazon Bedrock, Amazon Textract, prompt engineering, semantic search, ChromaDB, FAISS, RAGAS (retrieval and faithfulness evaluation), NLP (NLTK)

Orchestration & Data: Apache Airflow 3.x and 2.x, Astronomer, Astro CLI, dynamic task mapping, TaskFlow API, Assets, DAG Versioning, ETL/ELT, dbt, Cosmos (Airflow + dbt), Spark/PySpark (including Spark on Kubernetes), Snowflake, Databricks, PostgreSQL, OpenLineage, Apache Superset, Looker

Cloud & Infrastructure: AWS, GCP, Azure, Kubernetes, Docker, Terraform

Reliability & Delivery: release gating, RAG quality gates, data quality checks, pipeline observability and telemetry, automated regression testing, GitHub Actions, CircleCI, Azure DevOps

APIs & Languages: Python, SQL, Bash, Java, FastAPI, REST APIs, Streamlit, Slack integrations

PROFESSIONAL EXPERIENCE

Senior AI Data Engineer (Freelance)
Sep 2026 – Present

Self-employed | Remote, United States

- Deliver Apache Airflow consulting and AI data engineering through Topmate: pipeline design and review, RAG systems, and orchestration of AI workloads.
- Data and AI expert mentor at Ediso.ai: help learners build hands-on data and AI projects, troubleshoot issues with practical solutions, and answer questions across data engineering and AI.

Open Source Contributor, Apache Airflow
Mar 2026 – Present

The Apache Software Foundation | Remote

- Merged 7 pull requests into Apache Airflow (PR list), including a SparkKubernetesOperator change that passes the application name as an environment variable to driver and executor pods (#64015); other work covers dynamic task mapping, scheduler improvements, bug fixes and documentation.
- Named Apache Airflow Champion (Jan 2026) for contributions, PR reviews and issue triage; ranked Top 5 globally in Q2 2026 and #1 globally in Q3 2026.
- Take part in Airflow 3 feature development and governance discussions; authored the airflowctl issue template; review pull requests for Apache Airflow and Apache Spark.

Senior Software Engineer, Astro Observe (Data Platform)
Mar 2022 – Nov 2025

Astronomer | Bengaluru, India (Remote) and Parsippany, NJ

Joined as an SDET and moved into data platform and AI engineering. Finished as the sole engineer owning quality and release reliability for Astro Observe, Astronomer's managed Airflow observability product, and end-to-end owner of Airflow 3 regression testing, working with the Astro Build and Run teams on adoption.

- Built and deployed a RAG service (LangChain, ChromaDB, OpenAI) that summarizes CI test logs per release and answers queries posed through a CLI, improving issue resolution speed by ~40%.
- Architected cloud-native pipeline frameworks in Python (Airflow DAGs; boto3, azure-sdk, gcp-sdk, astro-sdk), increasing throughput ~30%.
- Tuned ETL/ELT orchestration on Airflow and Kubernetes across GCP, Azure and AWS, cutting data delivery time by 20%.
- Automated cloud resource lifecycle management with GCP and AWS CLI tooling, cutting infrastructure cost ~60%.
- Built the Airflow DAGs that drive release gating and sign-off for the Observability team, giving every team commit-level visibility into release status.
- Rebuilt the QA automation framework into the Astro release pipeline and embedded data quality and observability checks in CI/CD (CircleCI, GitHub Actions) to protect pipeline reliability at enterprise scale.
- Resolved UI and backend defects surfaced by automation framework runs.

Senior Test Automation Engineer, Core Payments Data May 2021 – Feb 2022
Commonwealth Bank of Australia

- Led automation strategy across a microservices architecture, reaching 95% coverage and cutting testing costs 15%; built reusable data validation frameworks for payment pipelines.

Consultant, Automation Specialist May 2020 – May 2021
KPMG India

- Built automation frameworks adopted by 4+ delivery teams, reducing manual QA cycles 25%; solely responsible for tool evaluation through proofs of concept across multiple vendors as part of the Center of Excellence.
- SME and day-to-day stakeholder contact for the Shell India account, including Terraform-based Azure automation and Azure Data Factory pipelines feeding Power BI.

IT Analyst / Software Engineer Jun 2014 – Apr 2020
Tata Consultancy Services

- Built CI/CD automation frameworks for Banking and Finance data platforms, cutting manual testing effort 50%; developed Java tooling for SQL and PL/SQL data operations, reducing manual database intervention 60%.

PROJECTS

- **Resume Analyser RAG** (LangChain, FAISS, OpenAI API, Python): multi-document retrieval and semantic search pipeline that answers natural-language questions over uploaded resumes. github.com/Shrividya/article-insights-pipeline/tree/main/rag_resume
- **Article Insights NLP Pipeline** (Apache Airflow 3, Python, NLTK): parallel DAG tasks for keyword extraction, summarization, sentiment analysis, entity recognition and readability scoring, with structured JSON output. github.com/Shrividya/article-insights-pipeline

SPEAKING, WRITING AND COMMUNITY

- **Speaker:** Airflow Summit 2026, Austin ("Orchestrating and Testing RAG Pipelines with Airflow"); Airflow Summit Online Reconnect, Nov 2026 ("Self-Healing Airflow: Pipelines That Fix Themselves"); Apache Airflow NYC Meetup, Jun 2026 ("Airflow + AI: Orchestrating the Intelligent Stack"); Future World Conversations; dbt community session on Airflow + dbt with Cosmos. Full talk history.
- **Judging:** Hackathon judge for Astronomer's "Beyond the Dag" hackathon and the Women in Data "Datathon".
- **Community:** Director, Women in Big Data, Atlanta Chapter (Oct 2026 – Present); Chapter Lead, Women in Data, New Jersey (Mar 2025 – Present), running technical bootcamps on SQL, Python, PySpark and applied ML; hosted Data Quality with Airflow sessions for Women in Data communities.
- **Mentoring:** Expert mentor for data and AI engineers on Ediso (Founding Expert Spotlight) and Topmate (5.0 average rating from 11 reviews).
- **Writing:** Medium and Substack articles, including RAG Without Vectors, Before You Deploy an AI Agent, Read This, What's Actually Costing You in Apache Airflow? and LangChain's Deep Agents Are Actually Worth Your Time. Editor for multiple tech publications on Medium; expert content writer for Data Engineering Bytes by Packt; editor for the IIP series, yearly volume 6S3 (2026).

EDUCATION AND CERTIFICATIONS

Education: Bachelor of Engineering, Computer Science and Engineering, Malnad College of Engineering, 2014 (CGPA 9.06 / 10)
Certifications: AWS Certified Cloud Practitioner (Jun 2025); Generative AI Fundamentals, Databricks Academy (Mar 2025); AI Orchestration with Apache Airflow, Astronomer (2026); DAG Authoring and Fundamentals for Airflow 3 (Jun 2025) and Airflow 2 (Mar – May 2025), Astronomer; AI Engineer for Developers Associate, DataCamp (Jul 2026)